

REAL ANALYSIS

1. Every function is a curve but every curve may not be a function.
2. $\text{Card}(\mathbb{Z}) = \text{Card}(\mathbb{N} \times \mathbb{N}) = \text{Card}(\mathbb{Q}) = \aleph_0$ (Aleph naught).
3. $\text{Card}(\mathbb{R}) = \text{Card}(\mathbb{R} \times \mathbb{R}) = c$ (Continuum).
4. There is a one-one correspondence $\mathbb{R} \rightarrow \mathbb{R} \times \mathbb{R}$.
5. There is a one-one correspondence between $\mathbb{R}^m$ and $\mathbb{R}^n$ for all positive integers m and n .
6. Every continuous open mapping of $\mathbb{R}$ to $\mathbb{R}$ is monotonic.
7. A metric space which is not complete is the space of all rational numbers, with $d(x, y) = |x - y|$.
8. Suppose $\{s_n\}$ is monotonic. Then $\{s_n\}$ converges iff it is bounded.
9. The product of two convergent series converges, and to the right value, if at least one of the two series converges absolutely.

10. For any set A ,
 - (i) $\text{int}(A) \subset A \subset \text{cl}(A) = A \cup A'$.
11. (i) If $A \subset B$, then $\text{int}(A) \subset \text{int}(B)$.
 (ii) $\text{int}(A) \cup \text{int}(B) \subset \text{int}(A \cup B)$.
 (iii) $\text{int}(A) \cap \text{int}(B) = \text{int}(A \cap B)$.
12. Suppose that $\{z_n\}$ is a complex sequence with limit z and suppose that $\{a_n\}$ is a positive sequence such that $\lim_{n \rightarrow \infty} (a_1 + a_2 + \cdots + a_n) = +\infty$.
 Then $\lim_{n \rightarrow \infty} \frac{a_1 z_1 + a_2 z_2 + \cdots + a_n z_n}{a_1 + a_2 + \cdots + a_n} = z$.
13. $\lim_{n \rightarrow \infty} \frac{x^n}{n^k} = +\infty$ if $x > 1$ and $k \in \mathbb{N}$.
14. $\lim_{n \rightarrow \infty} n^k x^n = 0$ if $|x| < 1$ and $k \in \mathbb{N}$.
15. $\lim_{n \rightarrow \infty} x^{\frac{1}{n}} = 1$ if $x > 0$.
16. For each $n \in \mathbb{N}$, $\lim_{k \rightarrow \infty} \frac{k^n}{2^k} = 0$.
17. For each $n \in \mathbb{N}$, $\lim_{k \rightarrow \infty} \frac{2^{nk}}{k!} = 0$.
18. For $m \geq 2$,
 - (i) $\frac{1}{k^m} \rightarrow 0$ as $m \rightarrow \infty$.
 - (ii) $\frac{1}{m^k} \rightarrow 0$ as $m \rightarrow \infty$.
19. Let $a_1, a_2, \dots, a_k \in \mathbb{R}^+$. Then $\lim_{n \rightarrow \infty} \sqrt[n]{a_1^n + a_2^n + \cdots + a_k^n} = \max\{a_1, a_2, \dots, a_k\}$.
20. Every increasing sequence in $\mathbb{Q}$ which is bounded above is a Cauchy sequence.
21. Every decreasing sequence in $\mathbb{Q}$ which is bounded below is a Cauchy sequence.
22. Let $\sum x_n$ be a series in $\mathbb{R}$ such that $x_n \geq 0$ for all $n \in \mathbb{N}$. Then $\sum x_n$ converges iff (s_n) is bounded.
23. If (x_n) is a decreasing sequence in $[0, \infty)$, then $\sum x_n$ converges iff $\sum 2^n x_{2^n}$ converges.

24. $\pi = \frac{1}{2} \min\{t > 0 \mid e^{it} = 1\}$.
25. Every Borel set is measurable set.
26. There exists measurable set which is not Borel set.
27. For any set A , there exists a measurable set E containing A (i.e., $E \subset A$) and such that $m^*(A) = m(E)$.
28. Every non-empty open set has positive measure.
29. There exists a non-measurable set.
30. $\log x = \int_1^x \frac{dy}{y}$.
31. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is one-one, then $d(a, b) = |f(a) - f(b)|$ is metric.
32. The directional derivative of $z = f(x, y)$ in the direction of unit vector $u = (a, b)$ is

$$D_u f(x, y) = \lim_{h \rightarrow 0} \frac{f(x + ah, y + bh) - f(x, y)}{h}$$

33. $D_u f(x, y) = \nabla f(x, y) \cdot u$
34. At a point (x, y) , the maximum value of the directional derivative $D_u f(x, y)$ is $|\nabla f(x, y)|$.
35. If $f(x, y)$ is differentiable, then f has a directional derivative in the direction of every unit vector $u = (a, b)$ and $D_u f(x, y) = f_x(x, y)a + f_y(x, y)b$.
36. Let $\{x_n\}$ be a sequence such that there exist $A > 0$ and $c \in (0, 1)$ for which $|x_{n+1} - x_n| \leq Ac^n$ for any $n \geq 1$. Then $\{x_n\}$ is Cauchy.
37. $\{p_n\}$ is a Cauchy sequence **iff** $\lim_{n \rightarrow \infty} \text{diam } E_N = 0$.
Where $E_N = \{p_N, p_{N+1}, \dots\}$ and $\text{diam } A = \sup\{d(x, y) \mid x, y \in A\}$
38. Let $\{f_n\}$ is the Fibonacci number sequence, then $f_n < 2^n$.
39. Test for continuity at $x = 0$
Left-hand limit $= \lim_{h \rightarrow 0} f(0 - h)$
Right-hand limit $= \lim_{h \rightarrow 0} f(0 + h)$

40. Test for differentiability at $x = 0$

$$\text{Left-hand derivative} = \lim_{h \rightarrow 0} \frac{f(0 - h) - f(0)}{-h}$$

$$\text{Right-hand derivative} = \lim_{h \rightarrow 0} \frac{f(0 + h) - f(0)}{h}$$

41. As $h \rightarrow 0$, $\sin(\frac{1}{h})$ oscillates between -1 and 1 .

42. As $h \rightarrow 0$, $\cos(\frac{1}{h})$ oscillates between -1 and 1 .

43. Let $f(x)$ be a periodic function. If $\lim_{x \rightarrow \infty} f(x)$ exists, then $f(x)$ is a constant.

44. $\lim_{x \rightarrow \infty} \sin x$ does not exist.

45. If $f''(x)$ exists and is continuous in a neighborhood of $x = a$, then

$$\lim_{h \rightarrow 0} \frac{f(a + h) - 2f(a) + f(a - h)}{h^2} = f''(a)$$

46. If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear map, then T is differentiable at every $a \in \mathbb{R}^n$ with $T'(a) = T(a)$.

47. If $f''(x) > 0$ for all $x \in \mathbb{R}$, then $f[\frac{1}{2}(x_1 + x_2)] \leq \frac{1}{2}[f(x_1) + f(x_2)]$.

48. Suppose $a_1 \geq a_2 \geq \cdots \geq 0$. Then the series $\sum_{n=1}^{\infty} a_n$ converges **iff** the series $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

49. The following functions are uniformly continuous

(i) $f : [1, \infty) \rightarrow \mathbb{R}$

(ii) Any linear map $T : \mathbb{R}^m \rightarrow \mathbb{R}^n$

(iii) $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f'(x)$ exists and is bounded.

50. The Arithmetic-Geometric Mean Inequality for a, b is $\sqrt{ab} \leq \frac{a+b}{2}$.

51. Let (x_n) be a sequence of real numbers and let $x \in \mathbb{R}$. If (a_n) is a sequence of positive real numbers with $\lim_{n \rightarrow \infty} (a_n) = 0$ and if for some constant $c > 0$ and some $m \in \mathbb{N}$, we have

$$|x_n - x| \leq ca_n, \quad n \geq m. \quad \text{Then } \lim_{n \rightarrow \infty} x_n = x.$$

52. If $0 < b < 1$, then $\lim_{n \rightarrow \infty} b^n = 0$.
53. If $c > 0$, then $\lim_{n \rightarrow \infty} c^{\frac{1}{n}} = 1$.
54. $\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$.
55. $\lim_{n \rightarrow \infty} \frac{\sin n}{n} = 0$.
56. Let (x_n) be a sequence of positive real numbers such that $l = \lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n}$ exists. If $|l| < 1$, then (x_n) converges and $\lim_{n \rightarrow \infty} (x_n) = 0$.
57. $x_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$ for $n \in \mathbb{N}$, is a divergent sequence.
58. $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e$, $2 < e < 3$
59. $\lim_{n \rightarrow \infty} (1 + \frac{a}{n})^{bn} = e^{ab}$.
60. $\lim_{x \rightarrow \infty} (1 + \frac{a}{x})^{bx} = e^{ab}$.
61. A continuous periodic function on $\mathbb{R}$ is bounded and uniformly continuous on $\mathbb{R}$.
62. If $f : A \rightarrow \mathbb{R}$ is a Lipschitz function, then f is uniformly continuous on A .
63. If $\lim_{|x| \rightarrow \infty} |f(x)| = 0$, then f is uniformly continuous.
64. If f has compact support, then f is uniformly continuous.
65. Suppose F is differentiable on I (interval) $|f'(x)| \leq M$, $\forall x \in I$, then F is uniformly continuous on I .
66. $\lim_{x \rightarrow 0} x \sin(\frac{1}{x}) = 0$.
67. Suppose $a \in [0, 1]$, then there exists sequences $x_n > 0$ and $y_n > 0$ such that $x_n \rightarrow 0$, $y_n \rightarrow 0$ and $x_n^{y_n} \rightarrow a$.
68. Suppose that $\phi : [0, \infty) \rightarrow \mathbb{R}$ is concave and $\phi(0) = 0$. Then

$$\phi(x + y) \leq \phi(x) + \phi(y), \quad \forall x, y > 0.$$

69. Let f be a continuous real valued function on $[0, \infty)$ such that $\lim_{x \rightarrow \infty} f(x)$ exists (finitely). Then f is uniformly continuous.
70. $f : \mathbb{R} \rightarrow \mathbb{R}$ continuous function and $|f(x) - f(y)| \geq |x - y|$ for all x and y . Then $f(\mathbb{R}) = \mathbb{R}$.
71. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a uniformly continuous function with the property that $\lim_{b \rightarrow \infty} \int_0^b f(x) dx$ exists (as a finite limit). Then $\lim_{x \rightarrow \infty} f(x) = 0$.
72. A real valued function of a complex variable either has derivative zero or the derivative does not exist.
73. Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be monotone on I . Then the set of points $D \subseteq I$ at which f is discontinuous is a countable set.
74. If $h : \mathbb{R} \rightarrow \mathbb{R}$ satisfies the identity $h(x + y) = h(x) + h(y)$, $\forall x, y \in \mathbb{R}$ and if h is continuous at a single point x_0 , then h is continuous at every point of $\mathbb{R}$.
75. If h is a monotone function satisfying $h(x + y) = h(x) + h(y)$, then h must be continuous on $\mathbb{R}$.
76. If f is bounded and there is a finite set E such that f is continuous at every point of $[a, b] \setminus E$, then f is Riemann integrable.
77. A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable **iff** it is continuous almost every where on $[a, b]$.
78. The only connected subsets of the real line are the intervals (open, closed, semi-open or semi-closed).
79. A non-empty open subset of the complex plane is connected **iff** any two of its points can be joined by polygon lying in the set having its segments parallel to the coordinates axes.
80. Every compact metric space is complete.
81. A subset S of a metric space X is compact **iff** S is complete and totally bounded.
82. A metric space is compact **iff** if every infinite sequence has a limit point.
83. Continuous image of connected set is connected.

84. Continuous image of a compact set is compact.
85. Suppose E is compact, f is continuous on E **iff** $\text{Graph}(f)$ is compact.
86. Let f be a differentiable real function on (a, b) . f is convex if and only if f' is monotonically increasing.
87. (X, τ) be a compact topological space and F be a closed subset of X . Then F is compact.
88. (X, τ) be a Hausdorff topological space and F be a compact subset of X . Then F is closed.
89. A complex sequence $\{z_n\}$ converges **iff** if it is Cauchy.
90. Suppose $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ ($x \in X$). Let $M_n = \sup_{x \in X} |f_n(x) - f(x)|$. Then $f_n \rightarrow f$ uniformly on X **iff** if $M_n \rightarrow 0$ as $n \rightarrow \infty$.
91. Suppose $\{f_n\}$ is a sequence of functions defined on X , and suppose $|f_n(x)| \leq M_n$ ($x \in X$, $n = 1, 2, \dots$). Then $\sum f_n$ converges uniformly on X if $\sum M_n$ converges.
92. The limit of uniformly convergent sequence of complex continuous functions on a metric space is continuous.
93. $f : X \rightarrow Y$ continuous. Then the graph of f with the subspace topology of $X \times Y$ is homeomorphic to X .
(i) $\mathbb{R}$ is homeomorphic to $y = x^2$. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$
94. Let $f : X \rightarrow Y$ be a homeomorphism and let $A \subset X$. Then f induces homeomorphism between A and $f(A)$, (and between $X \setminus A$ and $f(X \setminus A)$).
(i) $[0, 1)$ is not homeomorphic to $(0, 1)$, take $f = I$.
95. Any two open intervals of the real line are homeomorphic.
96. Any two closed intervals of the real line are homeomorphic.
97. Let $f : X \rightarrow Y$ be a continuous bijection. If X is compact and Y is Hausdorff then f is a homeomorphism.
98. A bijective continuous map is a homeomorphism **iff** it is open map.
99. A bijective continuous map is a homeomorphism **iff** it is closed map.

100. A bijective continuous map from a compact metric space to another metric space is a closed map and hence is a homeomorphism.
101. If $\sum c_n$ is convergent and $|\arg c_n| \leq \alpha < \frac{\pi}{2}$, then it converges absolutely.
102. If $\varphi : [a, b] \rightarrow \mathbb{R}$ is a step function, then $\varphi \in R[a, b]$
103. If $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$, then $f \in R[a, b]$
104. If $f : [a, b] \rightarrow \mathbb{R}$ is monotone on $[a, b]$, then $f \in R[a, b]$.
105. Two norms are equivalent
- (i) $\|x\|_2 \leq \|x\|_1 \leq \sqrt{n} \|x\|_2$
 - (ii) $\|x\|_\infty \leq \|x\|_2 \leq \sqrt{n} \|x\|_\infty$
 - (iii) $\|x\|_\infty \leq \|x\|_1 \leq n \|x\|_\infty$
 - (iv) $\frac{1}{d} \|x\|_1 \leq \frac{1}{\sqrt{d}} \|x\|_2 \leq \|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$
106. Box topology on $\mathbb{R}^w$
- (i) The box topology is completely regular.
 - (ii) The box topology is neither compact nor connected.
 - (iii) The box topology is not first countable.
 - (iv) Neither is the box topology separable.
 - (v) The box topology is para compact if the continuum hypothesis is true.
107. Every metric space is normal.
108. Every compact Hausdorff space is normal.
109. $(\{0, 1\}^{\mathbb{N}}, \text{box topology})$ is not compact.
110. The **orthonormal group**
 $O(n) = \{A \in \text{Mat}(n; \mathbb{R}) : AA^T = I\}$ is compact.
111. The **Special orthogonal group**
 $SO(n) = \{A \in O(n) : \det(A) = 1\}$ is a closed subgroup of the compact group $O(n)$, and so is itself compact.
112. The **Unitary group**
 $U(n) = \{A \in \text{Mat}(n; \mathbb{C}) : T^*T = I\}$ is compact.

113. The **Special unitary group**

$$SU(n) = \{A \in U(n) : \det(A) = 1\}$$

is a closed subgroup of the compact group $U(n)$, and so is itself compact.

$$114. U(n) = \frac{U(1) \times SU(n)}{\langle w \rangle}, \quad w = e^{2\pi/n}$$

115. The **Symplectic group**

$$Sp(n) = \{A \in Mat(n; H) : T^*T = I\}$$

(Here $Mat(n; H)$ denotes the space of $n \times n$ matrices with quaternian entries) is compact.

116. A matrix Lie group is any subgroup G of $GL(n; \mathbb{C})$ with the following property; If A_m is any sequence of matrices in G and A_m converges to some matrix A , then either $A \in G$ or A is not invertible. (i.e, a matrix Lie group is a closed subgroup of $GL(n; \mathbb{C})$)

117. The **complex orthogonal group**

$$O(n; \mathbb{C}) = \{A \in GL(n; \mathbb{C}) \mid \langle Ax, Ay \rangle = \langle x, y \rangle, \quad \forall x, y \in \mathbb{C}^n\}$$

.

118. The **Special complex orthogonal group**

$$SO(n; \mathbb{C}) = \{A \in O(n; \mathbb{C}) \mid \det(A) = 1\}$$

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119. Let g denote the $(n+k) \times (n+k)$ diagonal matrix with ones in the first n diagonal entries and minus ones in the last k diagonal entries.

$$O(n; k) = \{A_{(n+k) \times (n+k)} \mid A^T g A = g\}$$

120. A matrix Lie group G is said to be compact if the following two conditions are satisfied:

- a) If A_m is any sequence of matrices in G , and A_m converges to a matrix A , then A is in G .
- b) There exists a constant c such that for all $A \in G$, $|A_{ij}| \leq c$ for all $1 \leq i, j \leq n$.

121.

<u>Compact</u>	<u>Non – compact</u>
$O(n)$	$GL(n; \mathbb{R})$
$SO(n)$	$GL(n; \mathbb{C})$
$U(n)$	$SL(n; \mathbb{R})$
$SU(n)$	$SL(n; \mathbb{C})$
$Sp(n)$	$O(n; \mathbb{C})$
$S^1 \cong U(1)$	$SO(n; \mathbb{C})$
	$O(n; k)$
	$SO(n; k)$
	$Sp(n : \mathbb{R})$
	$Sp(n; \mathbb{C})$
	$\mathbb{R}, \mathbb{R}^n, \mathbb{R}^*, \mathbb{C}^*$

122.

<u>Group</u>	<u>Connected?</u>
$GL(n; \mathbb{C})$	<i>yes</i>
$SL(n; \mathbb{C})$	<i>yes</i>
$GL(n; \mathbb{R})$	<i>no</i>
$SL(n; \mathbb{R})$	<i>yes</i>
$O(n)$	<i>no</i>
$SO(n)$	<i>yes</i>
$SU(n)$	<i>yes</i>
$O(n; 1)$	<i>no</i>
$S0(n; 1)$	<i>no</i>

123. Let $f(x)$ and $g(x)$ be two differentiable functions with continuous n^{th} derivatives. Then

$$\frac{d^n}{dx^n}(f(x)g(x)) = \sum_{k=0}^n \binom{n}{k} \frac{d^k}{dx^k}(f(x)) \frac{d^{n-k}}{dx^{n-k}}(g(x)).$$

$$124. U(P, f) = \sum_{i=1}^n M_i \delta_i, \delta_i = x_i - x_{i-1}.$$

$$125. L(P, f) = \sum_{i=1}^n m_i \delta_i, \delta_i = x_i - x_{i-1}$$

$$126. \text{ For every partition } P \text{ on } [a, b], \text{ we have } L(P, f) \leq U(P, f)$$

$$127. m(b-a) \leq L(P, f) \leq U(P, f) \leq M(b-a)$$

$$\text{where } m = \inf\{f(x) \mid x \in [a, b]\}$$

$$M = \sup\{f(x) \mid x \in [a, b]\}$$

$$128. \int_a^{b-} f(x)dx = \inf\{U(P, f) : P \text{ is partition of } [a, b]\}$$

$$129. \int_{a-}^b f(x)dx = \sup\{L(P, f) : P \text{ is partition of } [a, b]\}$$

$$130. L(P, -f) = -U(P, f)$$

$$131. U(P, -f) = -L(P, f)$$

$$132. \int_{a-}^b f(x)dx \leq \int_a^{b-} f(x)dx$$

$$133. U(P, f) \leq \int_a^{b-} f(x)dx + \epsilon$$

$$134. L(P, f) \geq \int_{a-}^b f(x)dx - \epsilon$$

$$135. W(P, f) = U(P, f) - L(P, f)$$

$$136. \text{ If } f \text{ is Riemann integrable on } [a, b], \text{ then}$$

$$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{r=1}^n hf(a+rh), h = \frac{b-a}{n}$$

$$137. \text{ If } f \text{ is integrable on } [0, 1], \text{ then}$$

$$\int_0^1 f(x)dx = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right)$$

$$138. L(P, f) \leq \int_{a-}^b f(x)dx \leq \int_a^{b-} f(x)dx \leq U(P, f)$$

139. $m(b-a) \leq \int_a^b f(x)dx \leq M(b-a), b \geq a$

140. $m(b-a) \geq \int_a^b f(x)dx \geq M(b-a), b \leq a.$

141. $\int_a^b f(x)dx = \mu(b-a), m \leq \mu \leq M.$

142. $|\int_a^b f(x)d\alpha(x)| \leq [\alpha(b) - \alpha(a)] \max_{a \leq x \leq b} |f(x)|.$

143. Let $f(x)$ be a continuous function and $\alpha(x) \in C^1$ in $a \leq x \leq b$. Then

$$\int_a^b f(x)d\alpha(x) = \int_a^b f(x)\alpha'(x)dx.$$

144. A function $\alpha(x)$ is of bounded variation in an interval $a \leq x \leq b$ **iff** $\alpha(x) = \alpha_1(x) + \alpha_2(x)$, where $\alpha_1(x)$ is increasing and $\alpha_2(x)$ is decreasing in $a \leq x \leq b$.

145. Let $f(x)$ be an increasing function and $\alpha(x)$ is continuous function on $[a, b]$. Then

$$\int_a^b f(x)d\alpha(x) + \int_a^b \alpha(x)df(x) = \alpha(b)f(b) - \alpha(a)f(a).$$

146. Let $g(x)$ be a continuous function and $f(x)$ is an increasing function on $[a, b]$. Then

$$\int_a^b f(x)g(x)dx = f(a) \int_a^\xi g(x)dx + f(b) \int_\xi^b g(x)dx.$$

147. Let $g(x)$ be a continuous function, $f(x)$ be an increasing function and $f(x) \geq 0$ on $[a, b]$. Then

$$\int_a^b f(x)g(x)dx = f(b) \int_\xi^b g(x)dx, \quad a \leq \xi \leq b.$$

148. Suppose $\{s_n\}$ is monotonic. Then $\{s_n\}$ converges **iff** it is bounded.
149. The set of all limit points of a bounded sequence is a compact set (closed set).
150. $f_n \rightarrow 0$ **iff** $|f_n| \rightarrow 0$.
151. $f_n > 0$, $\forall n \in \mathbb{N}$, $\lim \frac{f_{n+1}}{f_n} = l$, $l > 0$, then $\lim (f_n)^{\frac{1}{n}} = l$.
152. If $|A| = m$, $|B| = n$, then the number of one-one functions from A to B is $\frac{n!}{(n-m)!}$ ($m \leq n$).
153. If $f(n)$ is a positive monotonically decreasing function of n for all $n \in \mathbb{N}$, then the two infinite series $\sum f(n)$ and $\sum a^n f(a^n)$ converge or diverge together, a being a positive integer greater than unity.
154. Let f be a function defined for all points in some neighbourhood of a point b except at the point b itself. Then $f(x) \rightarrow l$ as $x \rightarrow b$ **iff** the limit of the sequence $\langle f(x_n) \rangle$ exists and is equal to l for any sequence $\langle x_n \rangle$, $x_n \neq b$ for any $n \in \mathbb{N}$, converging to b .
155. If $f(x)$ has period T , then $af(bx + c) + d$ has period $\frac{T}{|b|}$.
156. If $f(x)$ and $g(x)$ has periods T_1 and T_2 , respectively, then $af(x) + bg(x)$ has period $T = lcm(T_1, T_2)$ provided exists.